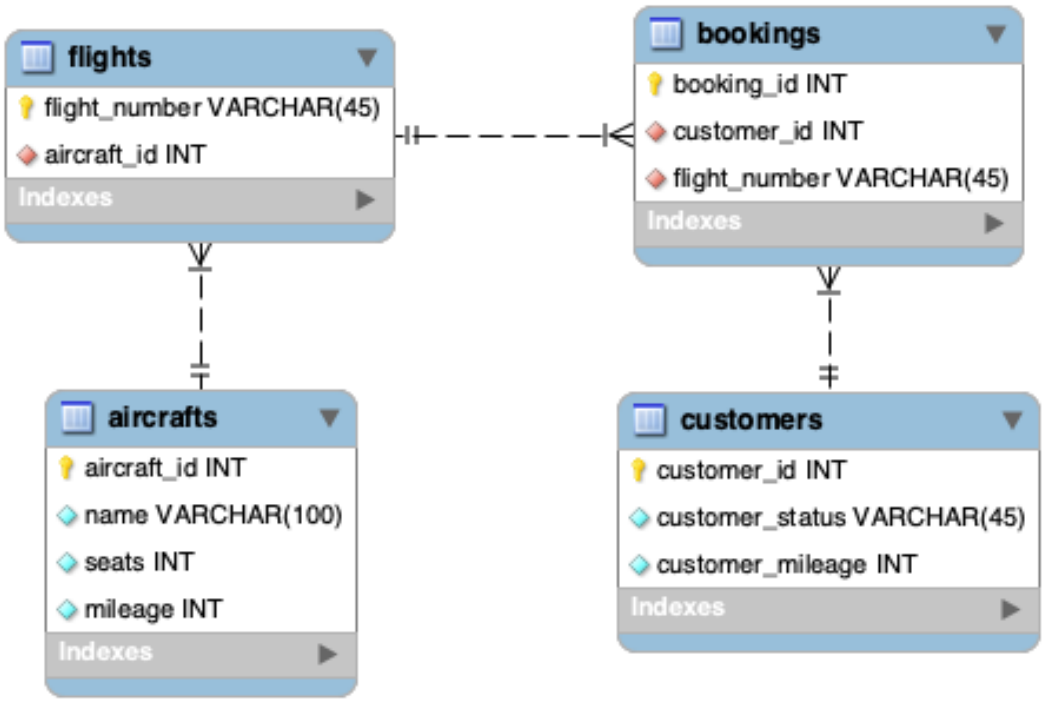
1. Normalize the following blog database and write the DDL scripts to create the database tables:

```
CREATE TABLE `author` (
  `author_id` int unsigned NOT NULL AUTO_INCREMENT,
  `author` varchar(45) NOT NULL,
  PRIMARY KEY (`author_id`)
);

CREATE TABLE `book` (
  `book_id` int unsigned NOT NULL AUTO_INCREMENT,
  `title` varchar(45) NOT NULL,
  `word_count` int NOT NULL,
  `views` int NOT NULL,
  PRIMARY KEY (`book_id`)
);
```

2. Normalize the following airline database and write the DDL scripts to create the database tables:



```
CREATE TABLE `aircrafts` (
  `aircraft_id` int unsigned NOT NULL AUTO_INCREMENT,
  `name` varchar(100) NOT NULL,
  `seats` int NOT NULL,
  `mileage` int NOT NULL,
  PRIMARY KEY (`aircraft_id`)
)

CREATE TABLE `bookings` (
  `booking_id` int NOT NULL AUTO_INCREMENT,
  `customer_id` int unsigned NOT NULL,
  `flight_number` varchar(45) NOT NULL,
  PRIMARY KEY (`booking_id`),
  KEY `customer_id` (`customer_id`),
  KEY `flight_number` (`flight_number`),
  CONSTRAINT `bookings_ibfk_1` FOREIGN KEY (`customer_id`)
REFERENCES `customers` (`customer_id`),
  CONSTRAINT `bookings_ibfk_2` FOREIGN KEY (`flight_number`)
REFERENCES `flights` (`flight_number`)
)

CREATE TABLE `customers` (
  `customer_id` int unsigned NOT NULL AUTO_INCREMENT,
  `customer_status` varchar(45) NOT NULL,
  `customer_mileage` int NOT NULL,
  PRIMARY KEY (`customer_id`)
)

CREATE TABLE `flights` (
  `flight_number` varchar(45) NOT NULL,
  `aircraft_id` int unsigned NOT NULL,
  PRIMARY KEY (`flight_number`),
  KEY `aircraft_id` (`aircraft_id`),
  CONSTRAINT `flights_ibfk_1` FOREIGN KEY (`aircraft_id`)
REFERENCES `aircrafts` (`aircraft_id`)
)
```

3. In the Airline database write the SQL script to get the total number of flights in the database.

```
SELECT COUNT(*) FROM flights;
```

COUNT(*)
5

4. In the Airline database write the SQL script to get the average flight distance.

```
SELECT AVG(mileage) FROM aircrafts;
```

AVG(mileage)
1775.8000

5. In the Airline database write the SQL script to get the average number of seats.

```
SELECT AVG(seats) FROM aircrafts;
```

AVG(seats)
312.8000

6. In the Airline database write the SQL script to get the average number of miles flown by customers grouped by status.

```
SELECT AVG(customer_mileage) FROM customers
GROUP BY customer_status;
```

AVG(customer_mileage)
98576.5000
4330.5000
253174.5000

7. In the Airline database write the SQL script to get the maximum number of miles flown by customers grouped by status.

```
SELECT MAX(customer_mileage) FROM customers
GROUP BY customer_status;
```

MAX(customer_milea...
136773
6008
300582

8. In the Airline database write the SQL script to get the total number of aircraft with a name containing Boeing.

```
SELECT COUNT(*) FROM aircrafts
WHERE name LIKE 'Boeing%';
```

COUNT(*)
4

9. In the Airline database write the SQL script to find all flights with a distance between 300 and 2000 miles.

```
SELECT flight_number FROM flights
INNER JOIN aircrafts ON aircrafts.aircraft_id =
flights.aircraft_id
WHERE mileage BETWEEN 300 AND 2000;
```

flight_number
DL222
DL37

10. In the Airline database write the SQL script to find the average flight distance booked grouped by customer status (this should require a join).

```
SELECT customer_status, AVG(mileage) FROM aircrafts
INNER JOIN flights ON aircrafts.aircraft_id =
flights.aircraft_id
INNER JOIN bookings ON flights.flight_number =
bookings.flight_number
INNER JOIN customers ON customers.customer_id =
bookings.customer_id
GROUP BY customer_status;
```

customer_status	AVG(mileage)
Silver	1438.3333
None	1292.7500
Gold	2494.5000

11. In the Airline database write the SQL script to find the most often booked aircraft by gold status members (this should require a join).

```
SELECT aircrafts.name, COUNT(*) AS total_bookings
FROM aircrafts
INNER JOIN flights ON aircrafts.aircraft_id =
flights.aircraft_id
INNER JOIN bookings ON flights.flight_number =
bookings.flight_number
INNER JOIN customers ON customers.customer_id =
bookings.customer_id
WHERE customers.customer_status = 'Gold'
GROUP BY aircrafts.name
ORDER BY total_bookings DESC
LIMIT 1;
```

name	total_bookin...
Boeing 777	3